Fəridə Mikayılzadə

KOM25

Linked list:

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```
class Node:
    def _init_(self, data):
        self.data = data
        self.next = None
class LinkedList:
    def _init_(self):
        self.head = None
    def insert_at_beginning(self, data):
        yeni_element = Node(data)
         yeni_element.next = self.head
        self.head =  yeni_element
    def insert_at_end(self, data):
         yeni_element = Node(data)
        if self.head is None:
            self.head =  yeni_element
            return
        son_element = self.head
        while son_element.next:
             son_element = son_element.next
    son_element.next =  yeni_element
    def delete_node(self, key):
        gecici = self.head

        if gecici is not None:
            if gecici.data == key:
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            self.head = gecici.next
            gecici = None
            return "Silindi"

        while gecici is not None:
            if gecici.data == key:
                break
            evvelki_node = gecici
            gecici = gecici.next

        if gecici == None:
            return "Element tapilmadi"

        evvelki_node.next = gecici.next
        gecici = None
        return "Silindi"

    def search_node(self, key):
        current = self.head
        while current:
            if current.data == key:
                return True
            current = current.next
        return False

    def reverse_list(self):
        evvelki = None
        current = self.head
        while current:
            novbeti = current.next
            current.next = evvelki
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            evvelki = current
            current = novbeti
        self.head = evvelki

    def print_list(self):
        elementler = []
        current = self.head
        while current:
            elementler.append(str(current.data))
            current = current.next
        print(" -> ".join(elementler) + " -> None")

if __name__ == "__main__":
    my_list = LinkedList()

    print("--- 1. Insert testleri ---")
    my_list.insert_at_beginning(10)
    my_list.insert_at_beginning(20)
    my_list.insert_at_end(30)
    my_list.insert_at_end(40)
    my_list.print_list()

    print("\n--- 2. Delete testi ---")
    my_list.delete_node(20)
    my_list.print_list()

    print("\n--- 3. Search testi ---")
    print("30 var mi?:", my_list.search_node(30))
    print("99 var mi?:", my_list.search_node(99))

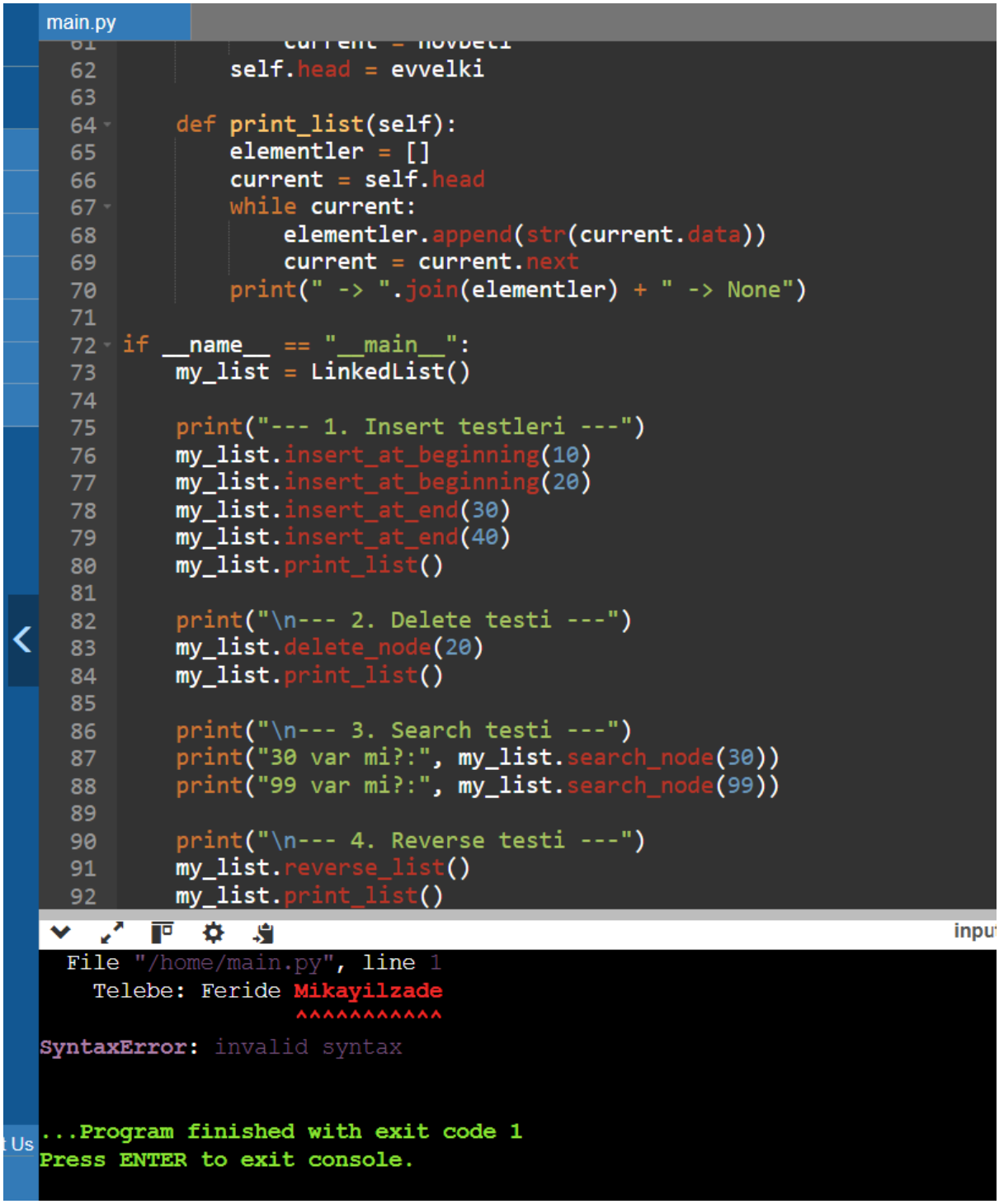
    print("\n--- 4. Reverse testi ---")
```

my_list.reverse_list()

my_list.print_list()

```
  File "/home/main.py", line 1
    Telebe: Feride Mikayilzade
                   ^^^^^^^^^^^
SyntaxError: invalid syntax


...Program finished with exit code 1
Press ENTER to exit console.
```



Bu tapşırıqda linked list struktru ilə tanış oldum, yeni biliklər qazandım və sadə şəkildə tətbiq etdim.